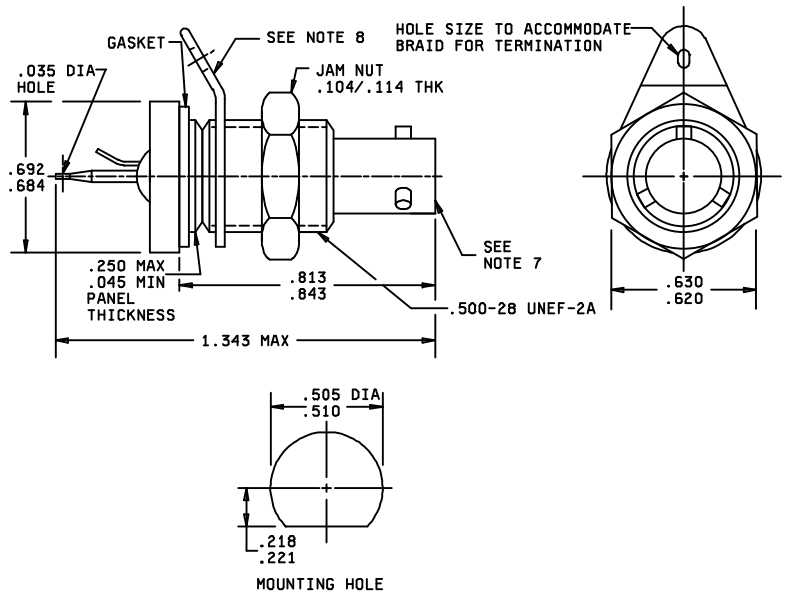
INCH-POUND
MIL-PRF-49142/4F
11 May 1998
SUPERSEDING
MIL-PRF-49142/4E
12 January 1990

PERFORMANCE SPECIFICATION

CONNECTORS, RECEPTACLE, ELECTRICAL, TRIAXIAL, RADIO FREQUENCY, UNCABLED, (SERIES TRB, SOCKET CONTACT, JAMNUT MOUNTED, CLASS 2) HERMETIC AND NONHERMETIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-49142.



Inches	mm
.022	.56
.035	.88
.045	1.14
.104	2.64
.114	2.90
.218	5.54
.221	5.61
.250	6.35
.500	12.70
.505	12.71
.510	12.95
.620	15.75
.630	16.00
.684	17.37
.692	17.58
.813	20.65
.843	21.41
1.343	34.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Wrench flats are to accommodate standard wrench openings in accordance with H-28, appendix 10.
4. All undimensioned pictorial representations are for reference purposes only.
5. Panel thickness, .045 (1.14 mm) minimum, .250 (6.35 mm) maximum.
6. Unless otherwise specified, all tolerances are ±.005 inch (0.13 mm).
7. Interface as specified in MIL-STD-348, series TRB, socket contact.
8. Only dash numbers -***6 and -***7 are supplied with braid termination lock washer rings. All previous dash numbers are supplied with standard lock washers (internal tooth .630 max. o.d. x .022 thick).

FIGURE 1. General configuration

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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ENGINEERING PARAMETERS:

Nominal impedance: Nonconstant.

Frequency range: 0 to 500 MHz, minimum.

Voltage rating: 400 V rms, maximum working voltage at sea level.
100 V rms, maximum working voltage at 70,00 feet.

Temperature range: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:
Longitudinal force: 4 pounds, maximum.
Torque: 2.5 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Mating characteristics: See figure 1 and MIL-STD-348 for dimensions.

Center contact (socket)
Oversize test pin: .040 inch diameter minimum (enclosed entry contacts only).
Insertion depth: .125 inch minimum.
Number of insertions: 1.

Insertion force test:
Steel test pin finish: 16 microinches.
Insertion force: 2 pounds, maximum.
Steel test pin diameter: .039 minimum +.001 inch.

Withdrawal force test:
Steel test pin diameter: .037 maximum -.001 inch.
Withdrawal force: 2 ounces minimum.
Test pin finish: 16 microinches.

Permeability: Applicable.

Hermetic seal: See table I.

Leakage: Connector shall be mounted in mounting hole shown on figure 1 with mating end capped.
Air pressure: 30 lbf/in/in.

Insulation resistance: 5000 megohms.

Conductor retention: 6 pounds, minimum, axial force.

Dielectric withstanding voltage: At sea level, 1200 V rms, between the center conductor and intermediate conductor.
500 V rms, between the intermediate conductor and the outer conductor.

Corrosion (salt spray): Applicable.

Vibration: Applicable.

Shock: Applicable.

Thermal shock: Applicable.

Moisture resistance: Applicable.

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Conductor resistance: Irmilliohms maximum.

	Connector dash number	Initial	After environment
Center conductor	All non-hermetic connectors	2.0	2.5
	All hermetic connectors	12.0	12.5
Intermediate conductor	All non-hermetic connectors	0.5	0.6
	All hermetic connectors	1.0	1.5
Outer conductor	All non-hermetic connectors	0.5	0.6
	All hermetic connectors	0.5	0.6

Corona level:
Altitude: 70,000 feet.
Voltage: 200 Vrms, minimum.

RF high potential withstanding voltage:
800 V rms, between center conductor and intermediate conductor.
200 V rms, between intermediate conductor and outer conductor at 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention: Not applicable.

Coupling mechanism: Not applicable.

Rise time degradation: Not applicable.

Connector durability: 500 cycles, minimum, at 12 cycles per minute, maximum.

Part or Identifying Number (PIN): M49142/04- (dash number from table I>)

Not for Naval Air Systems Command (AS) use.

TABLE I. Cross reference of PIN

Dash number	Superseded PIN
-0004 1/ -0006 1/ 2/ -0104, -01062/ -0204, -02062/ -3004, -30062/ -3104, -31062/ -3204, -32062/	M49142/04-0001, -0002 Non-hermetic
-0005 1/ -0007 1/ 2/ -0105, -01072/ -0205, -02072/ -3005, -30072/ -3105, -31072/ -3205, -32072/	M49142/04-0003 Hermetic

1/ Preferred keying.
2/ These connectors supplied with braid termination lock washer rings.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

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